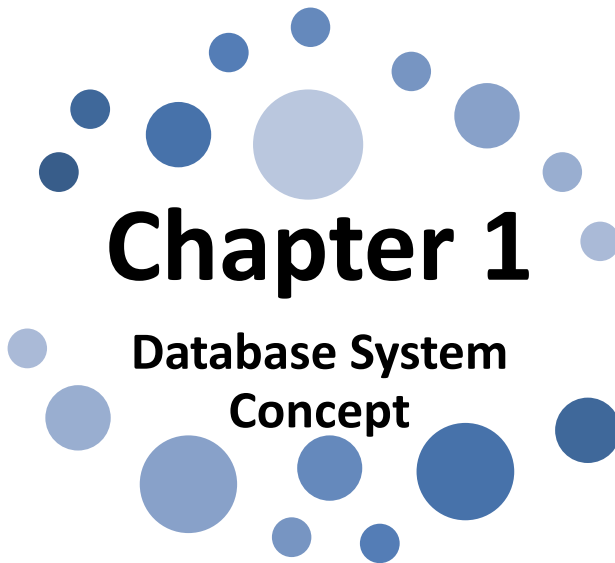




Chapter 1

Database System Concept

Unit 1.2: Three level Architecture for Database System

**Unit 1.3: Data Abstraction: Different Levels of Data
Abstraction, Instances and Schema, Data
Independence – Logical and Physical Independence**

Unit 1.4: Overall Structure of DBMS

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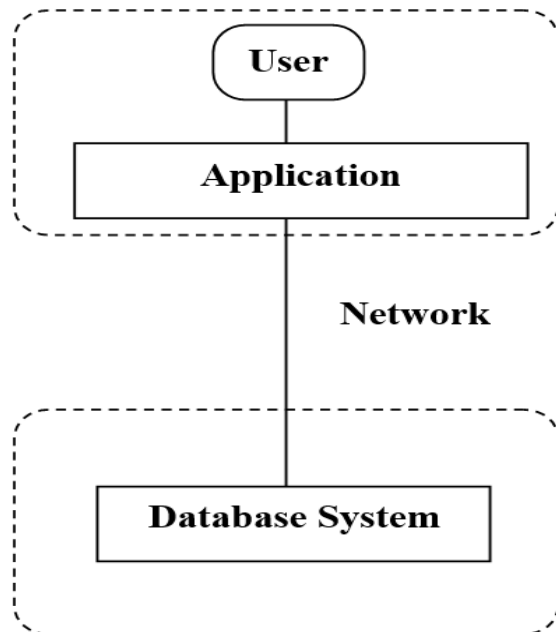
Learning Outcomes

The Learners will be able to:

1. **Understand** the Architecture of Database System.
2. **Understand** the concept of Data Abstraction.
3. **Describe** the overall structure of given DBMS.

Three level Architecture for Database System

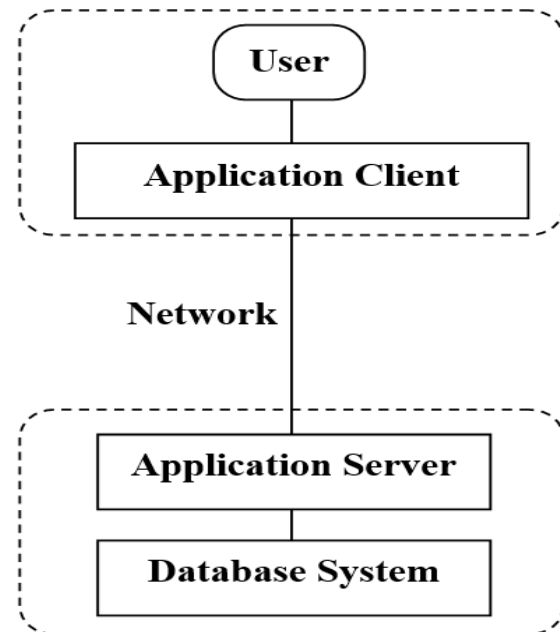
- This is an architecture of a computer network in which many clients (remote computers) request and receive information from a centralized server.
- Database architecture is logically divided into two types.
 1. **Two level Architecture OR two-tier Client / Server architecture**
 2. **Three level Architecture OR three-tier Client / Server architecture**



(a) Two-tier Architecture

client

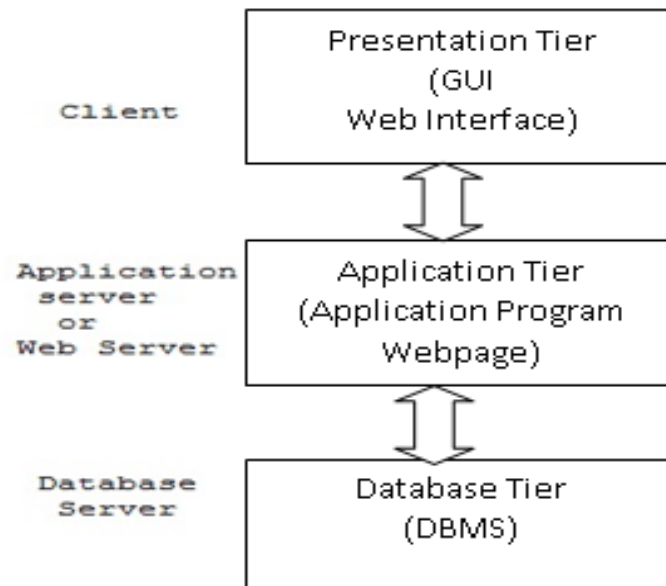
Server



(a) Three-tier Architecture

Three level Architecture for Database System

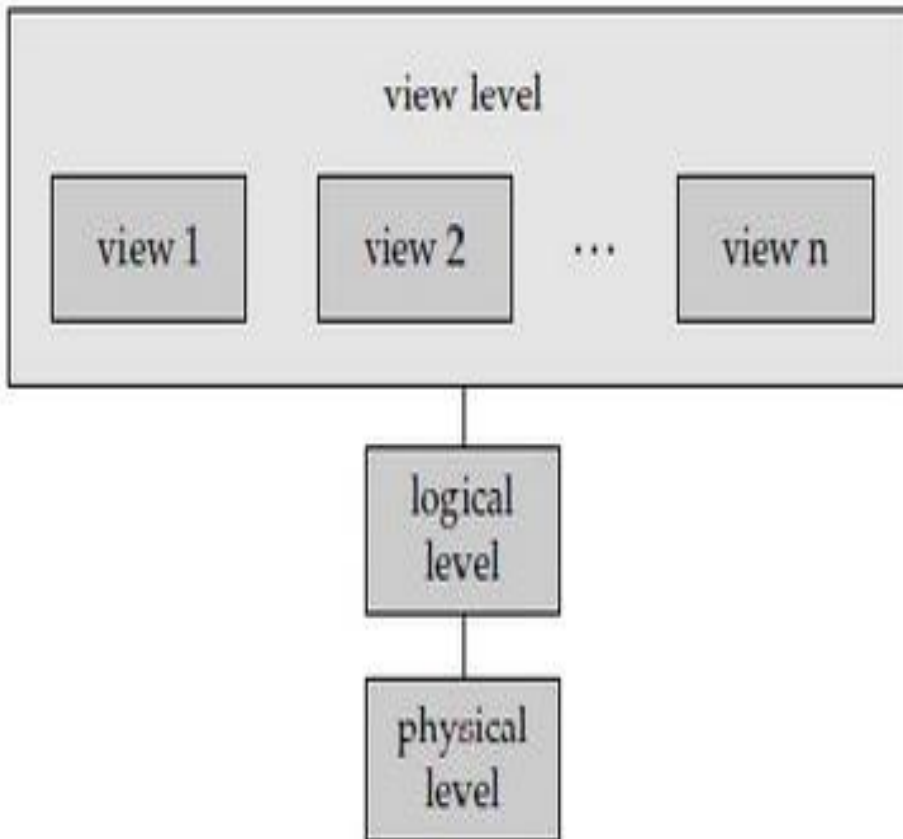
- Three level database architecture is commonly used for web applications.
- Intermediate layer called **Application server** or **Web Server** stores the web connectivity software and the logic(constraints).
- This layer acts like medium for sending partially processed data between the database server and the client.
- It separates its tiers from each other based on the complexity of the users and their access to database.
- It is the most widely used architecture to design a DBMS.



Data Abstraction

- The major purpose of a database system is to provide users with an **abstract view** of the system.
- The system hides certain details of how data is stored and created and maintained.
- Complexity should be hidden from database users.
- Hiding the details from one level to other level is known as **data abstraction**.
- There are several levels of data abstraction:
 - **View Level**
 - **Logical / Conceptual Level**
 - **Physical Level**

Different Level of Data Abstraction



What data,
users and
application
programs see?



What data is
stored?



How data
is actually
stored?

Different Level of Data Abstraction

View Level

- This is the highest level of abstraction.
- Many users need to access only a part of the database,
- Describe part of the database for a particular group of users.
- View level of abstraction simplify the user interaction with the database system and hide the complexities.
- The database system may provide many views for same database.

Logical Level

- This is the next higher level of abstraction.
- Describe what data are stored in the database.
- Describes what relationships exist among those data.
- Describes the entire database in terms of a small number of relatively simple structure.
- DBA use the logical level of abstraction.

Physical Level

- This is the lowest level of Abstraction.
- Describes how the data/records are actually stored
- The physical level describes complex low-level data structures in detail.

Reflection Spot

How will you distinguish between Logical Level and View Level?

Instances and Schema

- The overall design, logical structure or description of the database is known as **database schema** or simply **schema**.
- It is specified during database design, not expected to change frequently.
- It is analogous to information of a **variable** in a program
- A schema is plan of the database that gives the names of the entities and attributes and the relationship among them.
- It includes the definition of the database name, the record type and the components that make up the records.
- Alternatively, it is defined as a **framework** into which the values of the data items are fitted. The values fitted into the frame-work changes regularly but the format of schema remains the same.
- Generally, a schema can be partitioned into two categories, i.e.,
 - **Physical schema**: database design at the physical level
 - **Logical schema**: database design at the logical level

Database Schema for Online Book database

BOOK

ISBN	Book_title	Category	Price	Copyright_date	Year	Page_count	P_ID
------	------------	----------	-------	----------------	------	------------	------

PUBLISHER

P_ID	Pname	Address	State	Phone	Email_ID
------	-------	---------	-------	-------	----------

AUTHOR

A_ID	Aname	City	State	Zip	Phone	URL
------	-------	------	-------	-----	-------	-----

AUTHOR_BOOK

A_ID	ISBN
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REVIEW

R_ID	ISBN	Rating
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Instances and Schema

- A **subschema** is a subset of the schema having the same properties that a schema has.
- It identifies a subset of areas, sets, records, and data names defined in the database schema available to user sessions.
- The **subschema** allows the user to view only that part of the database that is of interest to him.
- The **subschema** defines the portion of the database as seen by the application programs and the application programs can have different view of data stored in the database.
- The different application programs can change their respective subschema without affecting other's subschema or view.

Instances and Schema

- The data in a database may change frequently, every time records are added or updated.
- The data in the database at a particular point of time is known as **database instance** or **database state** or **snapshot**.
- Many instances can correspond to a particular database schema.
- Every time we update (i.e., insert, delete or modify) the value of a data item in a record, one state of the database changes into another state.
- Analogous to the **value** of a **variable**

Two Instances of PUBLISHER schema

An example of instance - PUBLISHER schema

P_ID	Pname	Address	State	Phone	Email_id
P001	Hills Publications	12, Park street, Atlanta	Georgia	7134019	h_pub@hills.com
P002	Sunshine Publishers Ltd.	45, Second street, Newark	New Jersey	6548909	Null
P003	Bright Publications	123, Main street, Honolulu	Hawai	7678985	bright@bp.com
P004	Paramount Publishing House	789, Oak street, New York	New York	9254834	param_house@param.com

Another instance - PUBLISHER schema after deleting publisher with P_ID P002

P_ID	Pname	Address	State	Phone	Email_id
P001	Hills Publications	12, Park street, Atlanta	Georgia	7134019	h_pub@hills.com
P003	Bright Publications	123, Main street, Honolulu	Hawai	7678985	bright@bp.com
P004	Paramount Publishing House	789, Oak street, New York	New York	9254834	param_house@param.com

Data Independence – Logical and Physical Independence

- The capacity to change the schema at one level without having to change the schema at the next higher level is known as **data independence**.
- Data independence occurs because when the schema is changed at some level, the schema at the next higher level remains unchanged.
- Only the mapping between the levels is changed.
- In general, the interfaces between the various levels and components should be well defined so that changes in some parts do not seriously influence others
- Two types of Data Independence:
 - **Logical Data Independence**
 - **Physical Data Independence**

Data Independence – Logical and Physical Independence

Logical Data Independence

- The ability to change the logical schema without having to change the external schemas or application programs.
- When data is added or removed, only the view definition and the mappings need to be changed in the DBMS that support logical data independence.
- Applications depend on the logical schema.
- If the conceptual schema undergoes a logical reorganization, application programs that reference the external schema constructs must work as before

Physical Data Independence

- The ability to modify the physical schema without changing the logical schema.
- Physical file reorganization to improve performance results in a change to the internal schema. If the same data as before remains in the database, the logical schema should not change.
- For example, providing an access path to improve retrieval speed of section records by semester and year should not require a query to be changed, although it should become more efficient by utilizing the access path

Overall Structure of DBMS

